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> #Programmer Name: Peter Chapman  
> #Date: November 5, 2024  
  
> #This code reads in the Car Mileage Data Set. It then finds  
> #the variance and standard deviation for the miles per gallon,  
> #regardless of country. Finally, the variance and standard deviation  
> #are calculated for MPG of both countries separately.  
  
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```
> #reading in CSV data set to data frame  
> mpg<-read.csv("Car Mileage.csv")  
> mpg
```

Country MPG

```
1 United States 18  
2 United States 15  
3 United States 18  
4 United States 16  
5 United States 17  
6 United States 15  
7 United States 14  
8 United States 14  
9 United States 14  
10 United States 15  
11 United States 15  
12 United States 14  
13 United States 15  
14 United States 14  
15 United States 22  
16 United States 18  
17 United States 21
```

18 United States 21
19 United States 10
20 United States 10
21 United States 11
22 United States 9
23 United States 28
24 United States 25
25 United States 19
26 United States 16
27 United States 17
28 United States 19
29 Japan 24
30 Japan 27
31 Japan 27
32 Japan 25
33 Japan 31
34 Japan 35
35 Japan 24
36 Japan 19
37 Japan 28
38 Japan 23
39 Japan 27
40 Japan 20
41 Japan 22
42 Japan 18
43 Japan 20
44 Japan 31
45 Japan 32

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46    Japan 31
47    Japan 32
48    Japan 24
49    Japan 26
50    Japan 29
51    Japan 24
52    Japan 24
53    Japan 33
54    Japan 33
55    Japan 32
56    Japan 28
>

> #variance of miles per gallon, regardless of country
> mpg_var<-var(mpg$MPG)
> mpg_var
[1] 47.22825
>

> #standard deviation of miles per gallon, regardless of country
> mpg_sd<-sd(mpg$MPG)
> mpg_sd
[1] 6.872281
>

> #variance of miles per gallon for USA
> US_var_mpg<-with(mpg, var(MPG[Country=="United States"]))
> US_var_mpg
[1] 18.84656
>

> #standard deviation of miles per gallon for USA

```

```
> US_sd_mpg<-with(mpg, sd(MPG[Country=="United States"]))
```

```
> US_sd_mpg
```

```
[1] 4.341263
```

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```
> #variance of miles per gallon for Japan
```

```
> JP_var_mpg<-with(mpg, var(MPG[Country=="Japan"]))
```

```
> JP_var_mpg
```

```
[1] 22.12037
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```
> #standard deviation of miles per gallon for Japan
```

```
> JP_sd_mpg<-with(mpg, sd(MPG[Country=="Japan"]))
```

```
> JP_sd_mpg
```

```
[1] 4.70323
```

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